Suppose you decide to organize a Snakes and Ladders competition with n participants. In this competition, each game of Snakes and Ladders involves three players. After the game is finished, they are ranked first, second, and third. Each player may be involved in any (non-negative) number of games, and the number need not be equal among players.

At the end of the competition, m games have been played. You realize that you forgot to implement a proper rating system, and therefore decide to produce the overall ranking of all n players as you see fit. However, to avoid being too suspicious, if player A ranked better than player B in at least one game, then A must rank better than B in the overall ranking.

You are given the list of players and their rankings in each of the m games. Describe and analyze an algorithm that produces an overall ranking of the n players that is consistent with the individual game rankings, or correctly reports that no such ranking exists.

1. It is always good practice to come up with an example to make sure you understand the problem statement. Give an example competition with four distinct players and the outcome of three games. Propose an overall ranking of the players. Is it valid?

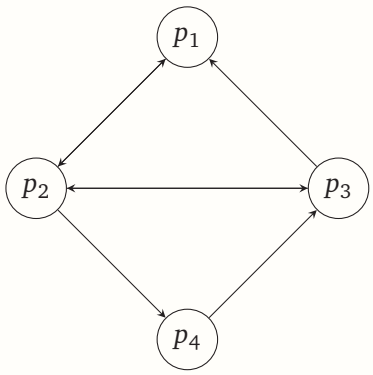
   Solution: Let's say the players are p_1, p_2, p_3, p_4, and they play the following games:

   - Game 1: p_2 first, p_3 second, p_1 third
   - Game 2: p_3 first, p_1 second, p_2 third
   - Game 3: p_2 first, p_4 second, p_3 third

   Let's ask whether p_1, p_2, p_3, p_4 is a valid ranking. We can see that the answer is ***no***, because p_4 ranked above p_3 in game 3, but p_4 does not rank better than p_3 in the overall ranking. ■

2. Your algorithm must be a reduction to a graph problem. Draw the graph you would build for your instance and use it to demonstrate that your overall ranking is or is not valid.

   Solution: The graph has one vertex for each player. For each game, we add a directed edge from every higher-ranked player to every lower-ranked player in that game:

   The graph contains the directed cycle $p_1 \to p_2 \to p_1$, so it has no topological ordering. Therefore, no overall ranking can satisfy all of the game constraints; in particular, the proposed ranking p_1, p_2, p_3, p_4 is not valid. ■

3. You are now ready to write down your full algorithm. You can do so by answering the following questions:

- What are the original parameters of the problem? (That is, the variables describing the input?) You will need this to analyze the runtime of your algorithm.

  Solution: n players and m games. ■

- What are the vertices? What does each vertex represent?

  Solution: There is one vertex per player. ■

- What are the edges? Are they directed or undirected?

  Solution: A directed edge exists between p_i and p_j if p_j ranked higher than p_i in some game. ■

- If the vertices and/or edges have associated values, what are they?

  Solution: None needed. ■

- **What is the precise relationship between your graph and the stated problem?**

  Solution: Any topological order of the vertices is a valid ranking, since no player can come before another in the toplogical order if they beat them in some game. If the graph has a cycle, then there is no toplogical order, and there is no valid ranking. ■

- What standard graph algorithm are you using to solve the problem?

  Solution: It would be okay to just say "check for cycles and compute a topological order", but to be more specific, we can run DFS recording post order values. If there is any edge $a \to b$ so that $a.post > b.post$, there is a cycle, and we can say there is no valid ranking. Otherwise, return the nodes in reverse post order. ■

- What is the running time of your entire algorithm, *including* the time to build the graph, *as a function of the original input parameters*?

  Solution: The graph has n vertices and $O(m)$ edges, since each game contributes up to 3 edges, so $V = n$ and $E = O(m)$. DFS takes $O(V + E)$ time and the brute-force check of edges takes $O(E)$ time. Returning a reverse post order of the vertices takes $O(V)$ time. Therefore, the overall runtime is $O(V + E)$, which is $O(n + m)$ in terms of the original parameters of the problem. ■